UNIVERSITÀ TELEMATICA INTERNAZIONALE UNINETTUNO

BERLIN SCHOOL OF BUSINESS & INNOVATION

Essay /Assignment Title: Innovation as a factor stimulating economic growth

Programme title: MSc in Engineering Management

Name: Baris Egemen Tokul

Year: 2025

CONTENTS

CONTENTS **2**
INTRODUCTION **4**
CHAPTER ONE **5**
Innovation as a Functional Process of Change 5
CHAPTER TWO **7**
Theory of Disruptive Innovation and Kotter's 8-Step Change Model 7
CHAPTER THREE **9**
Market Failures, Incentives, Market Structures, and Intellectual Property in Innovation Processes 9
CHAPTER FOUR **11**
Theoretical Analysis of the Economics of Innovation and Critical Assessment of Empirical Research 11
CONCLUDING REMARKS **14**
BIBLIOGRAPHY **16**

Statement of compliance with academic ethics and the avoidance of plagiarism

I honestly declare that this dissertation is entirely my own work and none of its part has been copied from printed or electronic sources, translated from foreign sources and reproduced from essays of other researchers or students. Wherever I have been based on ideas or other people texts I clearly declare it through the good use of references following academic ethics.

(In the case that is proved that part of the essay does not constitute an original work, but a copy of an already published essay or from another source, the student will be expelled permanently from the postgraduate program).

Name and Surname (Capital letters): BARIS EGEMEN TOKUL

.....

Date:15/10/2025

INTRODUCTION

Innovation has become one of the leading factors in the business and economic development of the digital age. A competitive advantage in modernity is no longer dependent on access to physical resources or cost-efficient production as it had been previously, but in a responsible and profitable business model, which allows the firm to innovate, evolve and transform (Schumpeter, 1934). In light of this increasing speed of communication, globalization and rapid expansion of technological development, innovation not only becomes an attribute of differentiation but of survival in the most competitive markets (Tidd and Bessant, 2020). Artificial intelligence (AI), disruption innovation and finance lead to a situation where new problems and possibilities occur for organisations, on the one hand AI makes it possible to decide quicker, to forecast and work more efficiently and it changes the parameters within which the firms can operate (Brynjolffson and McAfee, 2017). On the other hand these very forces which are involved here create turmoil in the traditional industries, produce new market leaders and expose firms to structural risks in their failure to adjust (Christensen, Raynor and McDonald, 2015). It is the dual role which innovation has in the modern world which this duality refers. Innovation is both a generator of process and a source of turbulence. The statement of the problem which defines this essay can be put in the following words: How can companies, not only adopt innovation, AI and financial solutions in achieving sustainable growth, but be adaptive to the systemic problems which afflict change? It is necessary in answering this question to consider how innovation is a transforming process, the disruptive nature of it and also the way firms finance and manage their growth through strategic means in dynamic situations. In this essay will be given the meanings and roles of innovation in business development first and the theory of disruptive innovation and the change management model expressed in 8-steps by Kotter will be subject to a critical examination. A case will be made of how AI is changing the nature of business development, a follow up of comment will be given on the scale-up and financial difficulties which firms involved in innovation have. The essay will conclude with evidence-based proposals to firms who feel the need of achieving a balance between the adoption of innovation, AI, management change and long-term financial planning.

CHAPTER ONE

Innovation as a Functional Process of Change

Innovation may be conceptualized not merely as the production of something new, but as an operative process of converting ideas, resources and capabilities into value creating outputs. In this regard, innovation is not a single event but rather a process that changes the manner in which firms, markets, and societies are being run (Schumpeter, 1934). It acts as an interface between the scientific or technological advancement and the practical implementation of the same in the realm of productivity, competitiveness and living standards. In essence, there are three key attributes to innovation:

1. Novelty - the presentation of new products, services, or processes that are not similar to the existing ones and possess the added value (OECD, 2018).
2. Implementation - conversion of creative concepts into practical solutions. An idea is not innovation until it is incorporated in the organizational systems, production systems, or market products (Tidd and Bessant, 2020).
3. Diffusion and Impact - this is the point at which innovation is economically relevant as it diffuses beyond the economic effect on a particular firm, to industries and societies.

This process is manifested through the large-scale introduction of mobile internet technologies or renewable energy technologies (Rogers, 2003). In the economic growth dimension, innovation is an impetus towards long-term growth. Whereas the classical models focused on accumulation of capital and labor productivity, contemporary economies are highly dependent on knowledge, innovation and flexibility (Solow, 1957). As an example, online platforms, such as Amazon and Alibaba, have not only developed new business models, but also transformed supply chains and world trade (Gawer and Cusumano, 2014). In this respect, innovation is a process of economic renovation itself a process of efficiency, the emergence of new industries and sometimes even the disruption of old ones. In such a way, the interpretation of innovation as a functional process of change is to recognize its duality: as an internal process of organizational change as well as external force, which reinvents markets and economies. In the contemporary days, where the

world is globalized, the pace of technology is quickly advancing and the expectations of the consumerism rapidly change, the process of innovation has ceased to be an option, it is the very process by which companies and countries will be preserved to be relevant and competitive.

CHAPTER TWO

Theory of Disruptive Innovation and Kotter's 8-Step Change Model

Disruptive Innovation Theory

The theory of disruptive innovation as proposed by Clayton Christensen offers one of the most powerful models of the way in which the markets are transformed by newcomers. Christensen argues that disruptive innovations are usually initiated in low-end parts or niche markets, whereby the existing companies do not consider them since they seem less lucrative or technologically insignificant (Christensen, Raynor and McDonald, 2015). With time, however, these innovations get better, increase their popularity, and ultimately, they replace established players. Disruptive innovation has an economic impact of distributing a market share and changing the cost structure. Incidentally, low-cost airlines have upset the conventional airlines by launching models that are low-cost but over time, the whole aviation industry transformed (Gans, 2016). In the same vein, the digital streaming applications posed a threat to the film and the music industry by exploiting the technology-driven distribution channels, lowering the cost of transactions, and changing consumer behavior. More importantly disruptive innovation is not merely about technological breakthrough. It is also powerful because of its business model innovation combined with new customer value propositions and scaleability. Companies that do not pay attention to disruptive entrants run the risk of losing relevance in the long-term in the market despite their seeming might in the immediate future (Markides, 2006).

8-Step Change Management Model by Kotter

Though the concept of disruptive innovation portrays the market level aspects of change, the 8-Step Model suggested by John Kotter provides an organized process of introduction of change in organizations. Kotter (1996) came up with eight step-by-step processes:

1. Creating a feeling of urgency.
2. Forming a guiding coalition

3. Setting up a vision and strategy.
4. Disseminating the vision of the change.
5. Strengthening the action of broad based action.
6. Generating short-term wins
7. Creating more change and putting it together.
8. Establishing new strategies within the culture.

The advantage of the model is that it is practical on leadership and communication by Kotter. In contrast to models with purely economic frameworks, Kotter recognizes social and behavioral resistance which usually cripples innovations. As an example, the employees might oppose the implementation of AI systems in case they see them as a threat to their employment. The stages of Kotter can enable managers to establish the legitimacy progressively, get the stakeholders in line, and assimilate the new practices into the organizational culture (Appelbaum et al., 2012).

The Two Frameworks Incorporated

Disruptive innovation theory and the Kotters change model are complementary in explaining innovation based transformation when viewed together. Christensen gives the reasons why disruption takes place and how companies need to predict changes in market structures, whereas Kotter gives the reasons as to how change may be handled within the organization in order to adapt to disruption. To provide an example, Netflix implementation of the streaming service instead of DVD rentals reflects the two frameworks. It was disruptive by capitalizing on digital delivery, which is the way it managed to change a whole industry, by bypassing the traditional distribution channels. Within the company, things were a bit different, but when it came to Kotter, it was still following some aspects of the model that included generating urgency, defining the vision, and entrenching the culture of constant innovation. Combination of these frameworks proves the fact that the successful innovation cannot be solely about how to identify the market changes but also how to implement the organizational change in a sound and engaging manner.

CHAPTER THREE

Market Failures, Incentives, Market Structures, and Intellectual Property in Innovation Processes

Innovation Market Failures

Innovation has been well known as a source of economic development, yet it has also experienced unique market failures, which is the reason why it needs to be intervened by the government. Among the failures is the knowledge spillover problem: when generated, knowledge can easily be copied with close to zero marginal costs meaning that innovators are not always able to fully reap the benefits of their investments (Arrow, 1962). It contributes to the insufficient investment in R&D, as social returns are usually higher than the private ones. The second failure is the uncertainty and risk. The cost of innovation and uncertainty are high sunk costs and uncertainty. One such area is the pharmaceutical development which is typified by enormous initial research spending with just a little percentage of the compounds being commercialized (Grabowski, Vernon & DiMasi, 2002). The capital markets can underestimate such uncertain returns and create financing gaps with innovative companies, particularly startups. Third, there is a failure in coordination when the complementary innovations are demanded. As an example, the effectiveness of electric cars is not only related to the automobile company, but also the charging network and battery technology providers. In the absence of coordination, the ecosystem might not attain critical mass (Teece, 2010).

Incentives in Firms in Dissimilar Markets

In spite of such failures, companies are encouraged to be innovative both strategically and defensively. Innovation can be used to lead the market in oligopoly markets. Companies invest to distinguish their goods and lose direct price competition, and thus, receive transitional monopoly payments (Aghion et al., 2005). Smaller firms tend to be more innovative to survive in competitive markets, with a focus on the niches that the incumbents are less efficient. Conversely, where there are monopolistic markets, then the dominant firms might not have strong incentives

to innovate due to the comfortable market position. This is referred to as the replacement effect where an entrant stands to capture less than an incumbent in terms of gains as a result of innovation (Reinganum, 1989). Nevertheless, in dynamic industries such as information technology, even large incumbents are under pressure to be defensive about innovation in case disruptive entrants come into the industry.

Innovation and its impacts on Markets

There are two impacts of innovation on market structures. On the one hand, it may enhance concentration where successful innovators may attain first-mover benefits, economies of scale, or network effects as in the case of platform companies like Amazon and Google (Shapiro and Varian, 1999). Conversely, innovation can also decrease barriers to entry and new companies could compete with the established ones. The growth of fintech startups represents an example of how consumer choice and competition are heightened by the reduced cost of entry into the financial services industry through digital technologies. In this way, the innovation may contribute, both, to facilitating creative destruction (Schumpeter, 1942) and reinforcing the leadership of the firms that manage to scale disruptive technologies successfully. The net effect is determined by industry attributes, regulation and incumbent flexibility.

Appropriability and Intellectual Property Rights (IPR)

In order to mitigate the spillover problem, intellectual property rights (IPR) focus on increasing appropriability, which is a temporary monopoly granted by the innovators. Patents offer protection under the law that encourages investment in the expensive R&D especially in pharmaceuticals, biotech, and clean technologies (Hall & Harhoff, 2012). Nevertheless, effective IPR regimes may create new inefficiency, including patent thickets which discourage cumulative innovation (Heller and Eisenberg, 1998). A portfolio of appropriability strategies is common in markets in which technology is changing at a rapid pace, including patents, secrecy, lead-time benefits, complementary resources, and brand name (Teece, 1986). To illustrate, Tesla employed patents at the beginning and then transferred to open-licensing strategy to hasten the development of the EV ecosystem, hoping to enjoy complementary benefits on infrastructure and brand leadership.

CHAPTER FOUR

Theoretical Analysis of the Economics of Innovation and Critical Assessment of Empirical Research

Theoretical Perspectives

Economics of innovation has emerged as an enriched field of research which integrates aspects of industrial organization theory, growth theory and evolutionary economics. The essence of the field is to research the role of innovation in improving productivity, competitiveness and long-term growth. The technological advances in neoclassical growth theory were initially exogenous (Solow, 1957). Here, innovation was a black box, which changed the production role upwards but was not attributed to firm-level decisions. This weakness prompted the emergence of the endogenous growth theory, where innovation is internalized as a result of intentionally created R&D and a cumulative knowledge (Romer, 1990; Aghion and Howitt, 1992). These models lay stress on the increasing returns to knowledge and the centrality of human capital hence offering a theoretical explanation of the policy of innovation. Another school of thought is evolutionary economics, which perceives innovation as a process of path-dependence, dynamic, routine-based, institution-based, and technology-dependent, (Nelson and winter, 1982). Contrary to the neoclassical models, evolutionary models emphasize the heterogeneity within firms and industries because innovation is not universal but rather context-specific. This opinion corresponds to the idea of creative destruction (Schumpeter, 1942) when the innovation breaks the current system and leads to the long-run industrial change.

Empirical Research and Evidence

These theories have been tested and even extended by empirical studies with mixed results. Among the sound results, there is the positive relationship between R&D intensity and productivity growth. Griliches (1992) proved that a company that invested more in R&D has a better performance in terms of output and market value as compared to its counterparts. The increasing number of cross-country studies since prove that R&D expenditure is associated with

total factor productivity, but the magnitude of the effect differs between institutional settings (Hall, Mairesse and Mohnen, 2010). The other body of research analyses the linkage between competition and innovation incentives. The study conducted by Aghion et al. (2005) found out that the relation between competition and incentives is inverted-U: insufficient competition kills incentives, and excessive competition kills the potential rent. This subtle opinion underpins the policies that would not eradicate the competitive pressure and preserve the profit possibilities. Nevertheless, later research has indicated that the curve shape varies between industries-pharmaceutical industries might need tighter patent protection, whereas digital markets might need to have the antitrust strengthened (Bloom, Draca & Van Reenen, 2016). Diffusion of innovation is also a subject of empirical research, and the diffusion theory presented by Rogers (2003) is greatly supported. Through empirical research, it has been established that adoption curves take S-shape, which is determined by network effects and complementary assets. However, the diffusion cycles of digital technologies are much quicker than the previous industrial inventions, which makes some people wonder whether the traditional models capture all the velocity of change (Comin & Hobijn, 2010).

Critical Assessment

Although the critical role of innovation in growth is mostly proven by empirical studies, there are a number of shortcomings. To begin with, there are measurement problems: innovation is not restricted by R&D expenditures or patents. Business model and organizational innovations are less measurable and this is dangerous to overlook their economic potential (OECD, 2018). Second, a lot of empirical work is situation specific. The experience of high-income economies can not be projected to the emerging markets where institutions, infrastructures and absorptive capacity are not the same (Cohen and Levinthal, 1990). Equally, the literature in the field of sectoral studies frequently points to paradoxical impact: IPR incentivizes pharma innovation, but can slow cumulative development in the field of software. Third, the social implications of innovation are increasingly being appreciated as necessitating a more subtle measure. Inequality, polarization of the labor market, and environmental sustainability are the issues that are increasingly discussed in empirical studies. As an example, Brynjolfsson and McAfee (2017) emphasize the role of digital innovation in improving productivity and causing job displacement, which contradicts the unconditionally positive welfare-promoting notion of innovation.

Synthesis

In theory, the economics of innovation has developed to exogenous models and evolutionary models; these models provide a more holistic view of the influence of innovation on the dynamics of economics. Empirically the contribution of innovation to growth is confirmed and complexes concerning the sector, institutions and society are revealed. More importantly, the field needs to keep on perfecting measurement instruments and its scope needs to go beyond the technological innovation and extend to the organizational, social, and environmental levels. It is only by this point that innovation economics can be used to give a holistic policy and business strategy in the digital era.

CONCLUDING REMARKS

This essay has discussed innovation as a key economic growth and organizational change agent in the digital era. With the understanding that innovation is a functional change, it became clear that the value of innovation is not just creating something new but also diffusing, impacting, and being competitive over time (Schumpeter, 1934; Tidd and Bessant, 2020). The comparison of the disruptive innovation theory and the 8-step Change Model created by Kotter revealed the dual nature of innovation as it both transforms the markets externally and needs planned change internally. Disruptive entrants have changed the cost structure and customer expectations but the successful change management allows organizations to surmount resistance and incorporate new practices into their culture (Christensen, Raynor and McDonald, 2015; Kotter, 1996). The presented evidence of the case study showed that artificial intelligence (AI) is a transformative force, which is essentially changing the manner in which businesses are conducted, decision-making, and interact with customers. The examples of Amazon and Tesla demonstrate that innovation and AI can help companies gain market dominance and transform the industry when the targeted actions are coordinated by the strategic vision and organized change management (Brynjolfsson and McAfee, 2017). Simultaneously the discussion of the market failures such as the knowledge spillovers, financing gaps, and the coordination problems demonstrated why market forces alone do not allow the innovation. The intellectual property rights, government policy, and mechanisms of complementary investments are key in maintaining incentives of innovation in industries (Arrow, 1962; Teece, 1986). Nevertheless, the threat of over-concentration or patent thickets indicates that the incentives have to be very carefully balanced with the larger social good when designing the policies (Heller and Eisenberg, 1998). Theoretical and empirical studies also proved the role of innovation in the center of growth and important details. The results of innovation vary in the industries, institutional environment, and social setting. In addition, more and more contemporary discussions acknowledge that innovation produces distribution effects, which affect the state of the labor markets, inequality, and environmental sustainability (Brynjolfsson and McAfee, 2017; OECD, 2018). To sum up, innovation is a chance and a difficulty. To promote sustainable development, companies have to make innovation their strategic core component, utilize the power of AI, and be capable of managing the change processes in an organization. In the meantime, policymakers have to devise

environments that promote innovation without having a systemic inefficiency. Eventually, how these forces can be balanced will lead to generating innovation as a way to achieve an inclusive, sustainable economic growth, or as a disruptive technology that will only increase inequalities and volatility.

BIBLIOGRAPHY

- Brynjolfsson, E. and McAfee, A. (2017) *Machine, Platform, Crowd: Harnessing Our Digital Future*. New York: W.W. Norton.
- Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', *Harvard Business Review*, 93(12), pp.44–53.
- Schumpeter, J.A. (1934) *The Theory of Economic Development*. Cambridge, MA: Harvard University Press.
- Tidd, J. and Bessant, J. (2020) *Managing Innovation: Integrating Technological, Market and Organizational Change*. 7th ed. Hoboken: Wiley.
- Gawer, A. and Cusumano, M.A. (2014) *Industry platforms and ecosystem innovation*. Journal of Product Innovation Management, 31(3), pp.417–433.
- OECD (2018) *Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation*. Paris: OECD Publishing.
- Rogers, E.M. (2003) *Diffusion of Innovations*. 5th ed. New York: Free Press.
- Schumpeter, J.A. (1934) *The Theory of Economic Development*. Cambridge, MA: Harvard University Press.
- Solow, R.M. (1957) 'Technical Change and the Aggregate Production Function', *Review of Economics and Statistics*, 39(3), pp. 312–320.
- Tidd, J. and Bessant, J. (2020) *Managing Innovation: Integrating Technological, Market and Organizational Change*. 7th ed. Hoboken: Wiley.
- Appelbaum, S.H., Habashy, S., Malo, J-L. and Shafiq, H. (2012) 'Back to the future: revisiting Kotter's 1996 change model', *Journal of Management Development*, 31(8), pp.764–782.
- Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', *Harvard Business Review*, 93(12), pp.44–53.
- Gans, J. (2016) *The Disruption Dilemma*. Cambridge, MA: MIT Press.
- Kotter, J.P. (1996) *Leading Change*. Boston: Harvard Business School Press.
- Markides, C. (2006) 'Disruptive innovation: In need of better theory', *Journal of Product Innovation Management*, 23(1), pp.19–25.
- Aghion, P., Bloom, N., Blundell, R., Griffith, R. and Howitt, P. (2005) 'Competition and innovation: An inverted-U relationship', *Quarterly Journal of Economics*, 120(2), pp.701–728.
- Arrow, K.J. (1962) 'Economic welfare and the allocation of resources for invention', in Nelson, R. (ed.) *The Rate and Direction of Inventive Activity*. Princeton: Princeton University Press, pp.609–626.
- Grabowski, H., Vernon, J. and DiMasi, J. (2002) 'Returns on research and development for 1990s new drug introductions', *Pharmacoeconomics*, 20(S3), pp.11–29.

- Hall, B.H. and Harhoff, D. (2012) 'Recent research on the economics of patents', *Annual Review of Economics*, 4, pp.541–565.
- Heller, M.A. and Eisenberg, R.S. (1998) 'Can patents deter innovation? The anticommons in biomedical research', *Science*, 280(5364), pp.698–701.
- Reinganum, J.F. (1989) 'The timing of innovation: Research, development, and diffusion', *Handbook of Industrial Organization*, 1, pp.849–908.
- Schumpeter, J.A. (1942) *Capitalism, Socialism and Democracy*. New York: Harper & Brothers.
- Shapiro, C. and Varian, H.R. (1999) *Information Rules: A Strategic Guide to the Network Economy*. Boston: Harvard Business School Press.
- Teece, D.J. (1986) 'Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy', *Research Policy*, 15(6), pp.285–305.
- Teece, D.J. (2010) 'Business models, business strategy and innovation', *Long Range Planning*, 43(2–3), pp.172–194.
- Aghion, P. and Howitt, P. (1992) 'A model of growth through creative destruction', *Econometrica*, 60(2), pp.323–351.
- Aghion, P., Bloom, N., Blundell, R., Griffith, R. and Howitt, P. (2005) 'Competition and innovation: An inverted-U relationship', *Quarterly Journal of Economics*, 120(2), pp.701–728.
- Bloom, N., Draca, M. and Van Reenen, J. (2016) 'Trade induced technical change? The impact of Chinese imports on innovation, IT and productivity', *Review of Economic Studies*, 83(1), pp.87–117.
- Brynjolfsson, E. and McAfee, A. (2017) *Machine, Platform, Crowd: Harnessing Our Digital Future*. New York: W.W. Norton.
- Cohen, W.M. and Levinthal, D.A. (1990) 'Absorptive capacity: A new perspective on learning and innovation', *Administrative Science Quarterly*, 35(1), pp.128–152.
- Comin, D. and Hobijn, B. (2010) 'An exploration of technology diffusion', *American Economic Review*, 100(5), pp.2031–2059.
- Griliches, Z. (1992) 'The search for R&D spillovers', *Scandinavian Journal of Economics*, 94(S), pp.29–47.
- Hall, B.H., Mairesse, J. and Mohnen, P. (2010) 'Measuring the returns to R&D', in Hall, B.H. and Rosenberg, N. (eds.) *Handbook of the Economics of Innovation*. Amsterdam: North-Holland, pp.1033–1082.
- Nelson, R.R. and Winter, S.G. (1982) *An Evolutionary Theory of Economic Change*. Cambridge, MA: Harvard University Press.
- OECD (2018) *Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation*. Paris: OECD Publishing.
- Romer, P.M. (1990) 'Endogenous technological change', *Journal of Political Economy*, 98(5), pp.S71–S102.

- Rogers, E.M. (2003) *Diffusion of Innovations*. 5th ed. New York: Free Press.
- Schumpeter, J.A. (1942) *Capitalism, Socialism and Democracy*. New York: Harper & Brothers.
- Solow, R.M. (1957) 'Technical change and the aggregate production function', *Review of Economics and Statistics*, 39(3), pp.312–320.
- Arrow, K.J. (1962) 'Economic welfare and the allocation of resources for invention', in Nelson, R. (ed.) *The Rate and Direction of Inventive Activity*. Princeton: Princeton University Press, pp.609–626.
- Brynjolfsson, E. and McAfee, A. (2017) *Machine, Platform, Crowd: Harnessing Our Digital Future*. New York: W.W. Norton.
- Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', *Harvard Business Review*, 93(12), pp.44–53.
- Heller, M.A. and Eisenberg, R.S. (1998) 'Can patents deter innovation? The anticommons in biomedical research', *Science*, 280(5364), pp.698–701.
- Kotter, J.P. (1996) *Leading Change*. Boston: Harvard Business School Press.
- OECD (2018) *Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation*. Paris: OECD Publishing.
- Schumpeter, J.A. (1934) *The Theory of Economic Development*. Cambridge, MA: Harvard University Press.
- Teece, D.J. (1986) 'Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy', *Research Policy*, 15(6), pp.285–305.
- Tidd, J. and Bessant, J. (2020) *Managing Innovation: Integrating Technological, Market and Organizational Change*. 7th ed. Hoboken: Wiley.

APPENDIX (if necessary)